

Organisms and
their environment

Food chain: Shows the transfer of energy from one organism to the next, starting with the producer.

Food web: A network of interconnected food chains

Producer: An organism that makes its own organic nutrients, using energy from sunlight, through photosynthesis.

Consumer: An organism that gets its energy by feeding on other organisms.

Herbivore: An animal that gets energy by eating plants

Carnivore: An animal that gets energy by eating other animals

Decomposer: An organism that gets its energy from dead or waste organic material.

Trophic layer: The position of an organism in a food chain, food web or ecological pyramid

Population: A group of organisms of one species, living in the same area at the same time

Community: All populations of different species in an ecosystem

Ecosystem: A unit containing the community of organisms and their environment, interacting together

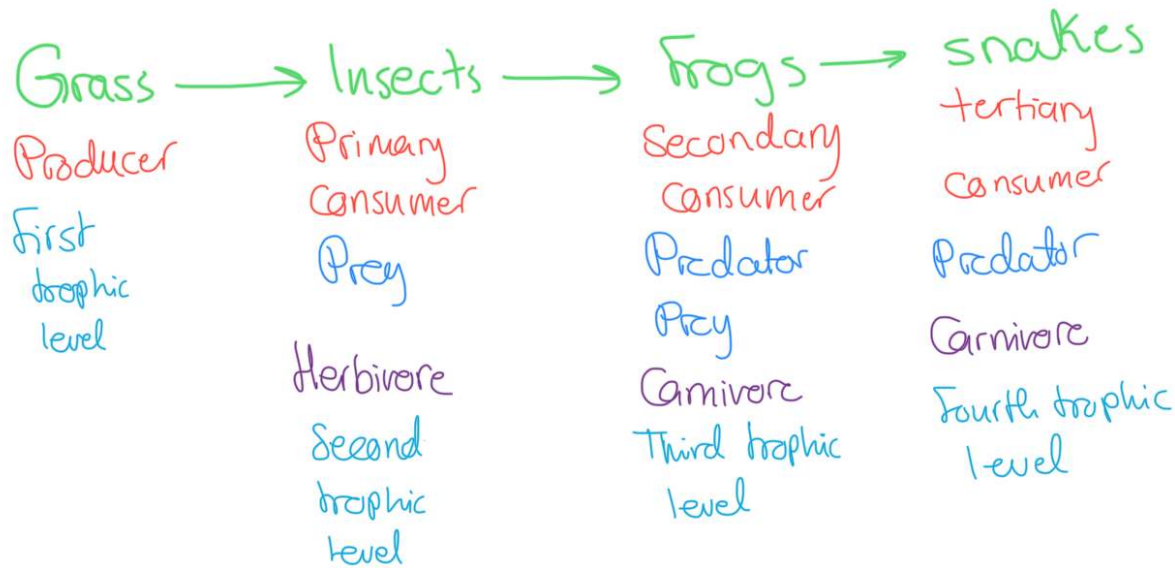
Fitness: The probability of an organism surviving and reproducing in the environment in which it is found.

* The sun is the principal source of energy into biological system

Why all food chains are not more than five trophic levels?

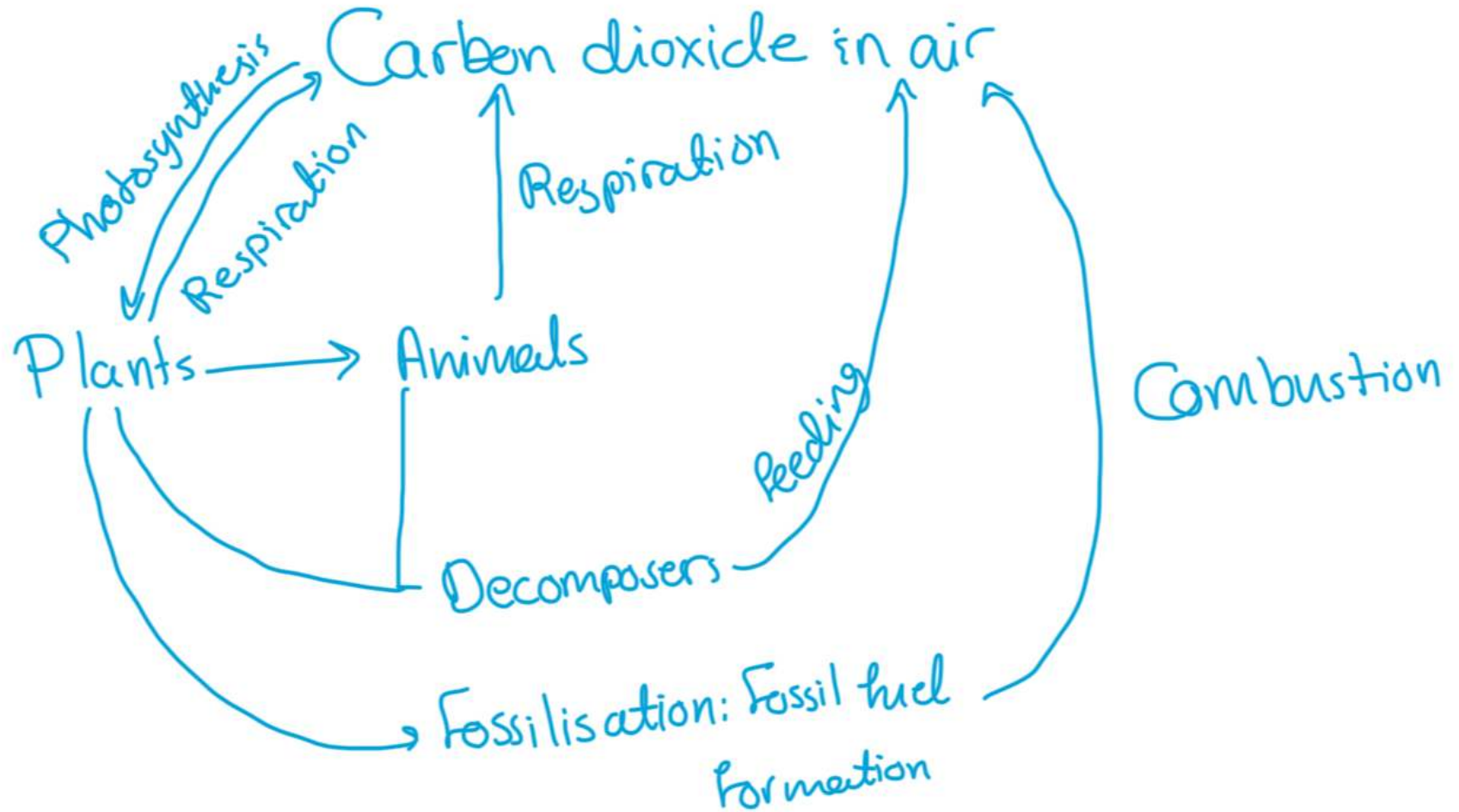
Because energy is lost between trophic levels in food chain.

Energy is lost by respiration, movement excretion. Not all organisms are eaten and digested. So there isn't enough energy left to support higher trophic levels

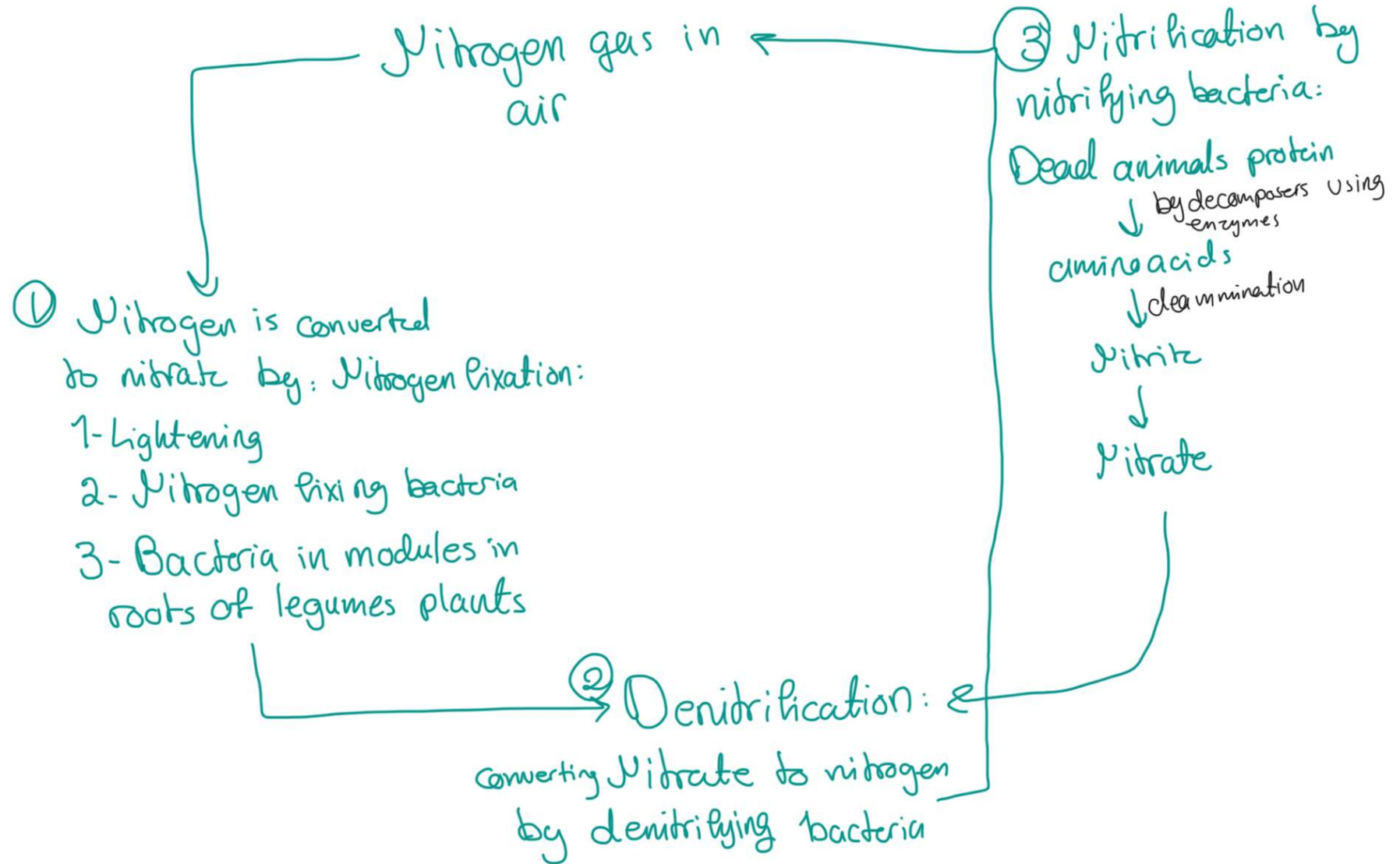


Efficiency of energy transfer → $\frac{\text{Energy in higher trophic level}}{\text{Energy in lower trophic level}} \times 100\%$

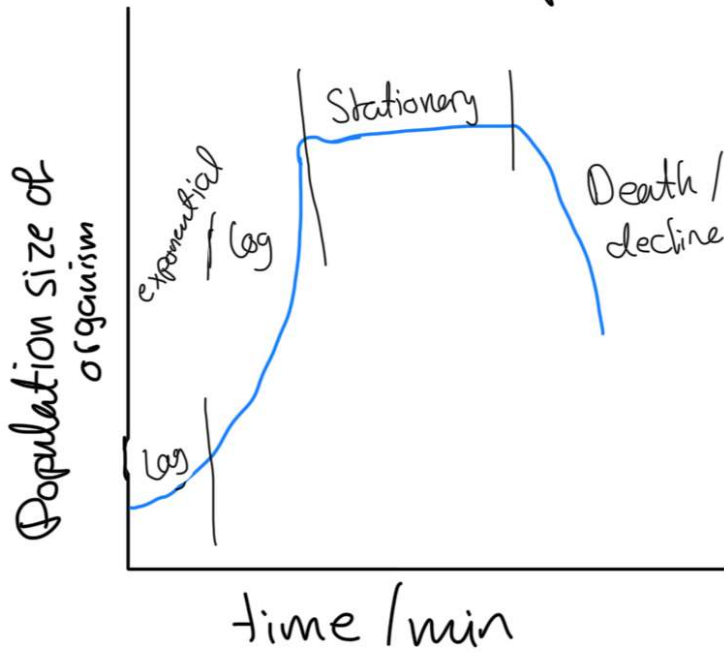
Carbon Cycle



Nitrogen Cycle



Population size:



Sigmoid curve

Lag phase:

Description: Little increase in population size

Explain: Living organism take time to adapt to new conditions

Log phase / exponential:

Description: Sharp increase in population size

Explain: Nutrients and space for growth are available. No competition
No limiting factor.

Birth rate $>$ Death rate

Stationary:

Description: No change in population size

Explain: Birth rate = Death rate
Shortage of nutrients, which is now limiting factor. Competition

Death / Decline:

Description: Decrease in population size

Explain: No nutrients, no space
build up of waste - Competition

Birth rate $<$ Death rate